

Theta Closure in Modal Triplet Theory I: Gauge-Profile Targets from Multi-Loop Common-Scheme Transport

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Abstract

We present a conditional Θ -closure realization relating Standard Model gauge couplings to internal geometric overlap integrals. Measured source data are transported with SMDR v1.3 into the full-Standard-Model $\overline{\text{MS}}$ scheme at $Q = M_t$. In the normalization used here, the resulting profile targets are

$$I_2/I_1 = 0.5110273 \pm 0.0001231, \quad I_3/I_1 = 0.158335 \pm 0.001098.$$

They are calibrated profile coordinates, not first-principles predictions of the gauge couplings. We formulate the admissibility and spectral-gap conditions under which an internal realization may reproduce these targets. To address the normalization of nonabelian harmonics, we introduce a high-coherence (twistor) corner with declared period and gauge-kinetic normalizations. An $O(\lambda_Q^{-1})$ overlap estimate is conditional on explicit projector and representative perturbation bounds. The scale $Q = M_t$ is a renormalization and matching convention, not a physical coherence, gap, or quantum-gravity scale. The result is a calibrated target and conditional existence test; independent geometric selection remains open.

Version 3 Revision Note

Supersedes. Version 2 of this paper; released identifiers are retained.

Reason. The calibration boundary was present, but the selected-source owner results and their distinct scopes were not explicitly connected to it.

Resolution. Version 3 imports the exact Gate 1, three-cycle and large-gauge-kernel conclusions through their owners, and also removes residual equality/ontology overstatements in the gap discussion.

Retained. The existing calibrated or conditional result, its numerical inputs and all earlier revision notes are retained.

Open boundary. A selected physical source, its action normalization and the paper-specific execution inputs remain necessary for a held-out prediction.

Revision note for this edition

Supersedes. *Theta Closure in Modal Triplet Theory I: Gauge Couplings from Internal Geometry*, first edition.

Reason. The former target used an obsolete one-loop few-TeV crossing, treated calibrated ratios as predictions, and combined unit- L^2 normalization with overlap definitions that would force trivial unit values.

Resolution. Version 2 uses weighted gauge-kinetic coefficients and SMDR v1.3 common-scheme targets at $Q = M_t$, with explicit provenance and conditional projector-error assumptions.

Retained result. Gauge-coupling ratios can still be posed as geometric overlap targets in a normalized high-coherence realization.

Remaining boundary. MTT must independently select and execute the internal geometry before these calibrated targets become predictions.

1 Purpose and scope

This paper presents a conditional map from a common-scheme Standard Model gauge profile to internal overlap targets in Modal Triplet Theory (MTT).

The goals are deliberately narrow:

1. Derive gauge-coupling expressions from internal overlap integrals.
2. Extract numerical Θ -profile targets from measured couplings.
3. Test whether a nonempty admissible Θ -region satisfies the MTT gap conditions.
4. State all assumptions and calibration choices explicitly.

No claim of uniqueness, zero-knob gauge-coupling prediction, or full unification is made. The coupling data enter upstream of the profile targets; agreement of a fitted geometry with those same targets is a realization or round-trip test, not a held-out prediction.

1.1 The central picture

In plain language, each gauge field samples an internal harmonic profile. Its four-dimensional kinetic coefficient is the weighted cost of that profile over the internal carrier. A larger coefficient I_a means a smaller four-dimensional coupling g_a when the higher-dimensional normalization is held fixed. The two ratios in this paper therefore act like two calibrated rulers: they specify how the weak and color profile costs compare with the abelian one, but they do not yet explain why MTT selected those costs.

The shared circle has a precise but limited role in this picture. It supplies common phase or holonomy data used by all three lanes and is counted once. The lane labels organize operators and representatives; they are not three independent compact spaces whose dimensions should be added.

1.2 How to read the dependency chain

The argument has four stages:

1. reduce a declared higher-dimensional gauge action to the relation $1/g_a^2 = I_a/g_{10}^2$;
2. transport measured gauge data into one scale and one renormalization scheme;
3. convert the transported coupling ratios into overlap-ratio targets;
4. ask whether an auxiliary internal representative can realize those targets while preserving the declared spectral gap.

Only the first and fourth stages concern the geometric realization. The second and third stages calibrate it from measured Standard Model data. This ordering is why existence of a matching representative is weaker than a prediction of the couplings.

2 Standing assumptions of Modal Triplet Theory

We collect here the assumptions used throughout. They are standard within the MTT corpus and are not re-derived.

Assumption 2.1 (Bounded internal geometry). The internal bundles B_n possess bounded geometry and admit commuting vertical Laplacians.

Assumption 2.2 (Uniform spectral gap). There exists a strictly positive lower bound

$$\lambda_* > 0$$

separating coherent from noncoherent internal modes.

Assumption 2.3 (Bounded coherent projection). The joint Riesz projector

$$\Pi = \Pi_{B_1} \Pi_{B_2} \Pi_{B_3}$$

is bounded on H^1 .

Assumption 2.4 (Fundamental Contractivity Condition (FCC)). The projected flow $F = \Pi \circ \Phi_\tau$ is contractive on the coherent sector.

Remark 2.5. If these assumptions fail, this paper loses controlled access to the stated coherent effective encoding. No claim is made that the underlying mathematical configuration, probabilities, records, or observers cease to exist in every description.

2.1 Spectral gap as a controlled-description gate

A uniform spectral gap is assumed throughout this work as a sufficient condition for a stable finite coherent sector. If the selected cluster loses separation, the corresponding Riesz projector and truncation estimates may lose regularity, so this paper no longer controls the stated effective encoding. This does not imply that amplitudes, probabilities, records, or an underlying mathematical description cease to exist in every formulation.

The present work does not derive the gap dynamically or prove that every physical theory must possess this particular finite-cluster description. It tests consequences conditional on the declared gap and projector hypotheses.

3 Internal operator structure

The physical dimensional-reduction formula is posed on one compact six-manifold X_6 with three compatible vertical sectors. A single Hermitian line bundle or principal $U(1)$ bundle $L_{\text{shared}} \rightarrow X_6$ supplies the common phase/holonomy datum. It acts in every sector and is counted once.

For the calibrated Route-A computation only, write B_n for the effective support and measure used by sector n . Locally these supports may be modeled using:

- a shared-circle phase normalization for the abelian lane,
- a round two-dimensional lens-base representative for the weak lane,
- a compact Heisenberg nilmanifold representative for the color lane.

The three B_n are not asserted to be simultaneous Cartesian factors of X_6 , and no literal $S^1_{\text{shared}} \times L(3, 1) \times \text{Nil}_3$ compactification is used. The Lens-Nil data form an auxiliary calibrated operator model; the current global physical candidate is the distinct q79/Fu-Yau branch.

4 Gauge couplings from internal overlaps

4.1 Definition of gauge-kinetic overlaps

Definition 4.1 (Sector gauge-kinetic coefficient). For each gauge factor $a = 1, 2, 3$, fix a sector representative ω_a by a declared pointwise, period, or cohomological normalization and define

$$I_a := \int_{B_a} w_a \langle \omega_a, \omega_a \rangle d\mu_{B_a}.$$

Here $w_a > 0$ contains the induced gauge-kinetic density, including any integrated spectator directions and trace factors. The representative is not simultaneously normalized to unit $L^2(B_a, w_a d\mu)$; doing so would make $I_a = 1$ by definition and erase the coupling information.

4.2 Reduction of the Yang–Mills action

Proposition 4.2 (Conditional coupling–overlap relation). *Assume a ten-dimensional Yang–Mills sector with common coefficient g_{10} , a consistent truncation to mutually orthogonal four-dimensional gauge zero modes, and the gauge-kinetic weights and normalizations in the preceding definition. Then*

$$\frac{1}{g_a^2} = \frac{1}{g_{10}^2} I_a.$$

Proof. Start from the ten-dimensional Yang–Mills action

$$S_{10} = \frac{1}{2g_{10}^2} \int \text{Tr}(F_{AB}F^{AB}) d\text{vol}_{10}.$$

Expand the gauge field along the declared internal sector modes

$$A_\mu(x, u) = A_\mu^{(a)}(x) \omega_a(u),$$

Integrating the weighted internal gauge-kinetic density and using orthogonality of the retained modes yields

$$S_4 = \frac{1}{4g_a^2} \int F_{\mu\nu}^{(a)} F^{(a)\mu\nu} d\text{vol}_4,$$

with

$$\frac{1}{g_a^2} = \frac{1}{g_{10}^2} \int_{B_a} w_a \langle \omega_a, \omega_a \rangle d\mu_{B_a}.$$

□

4.3 Normalization conventions

We adopt the following conventions:

1. **Generator normalization:**

$$\text{Tr}(T^a T^b) = \frac{1}{2} \delta^{ab}.$$

2. **Hypercharge normalization (GUT-normalized):**

$$g_1 = \sqrt{\frac{5}{3}} g', \quad \alpha_1 = \frac{5}{3} \alpha_Y.$$

With these conventions, the normalization constants satisfy

$$N_1 = N_2 = N_3 = 1,$$

and all group-theoretic factors are absorbed into the definition of g_1 .

5 Summary of this section

At this point we have established:

- Gauge couplings descend from weighted internal gauge-kinetic coefficients under the stated higher-dimensional action and truncation assumptions.
- The relation $1/g_a^2 = I_a/g_{10}^2$ is exact within that declared reduction ansatz.
- Ratios of couplings are ratios of those coefficients only when the same g_{10} and compatible generator conventions apply to all three sectors.

In the next part we will:

1. Compute g_1, g_2, g_3 explicitly from experimental inputs.
2. Run them to a common matching scale.
3. Extract numerical Θ -constraints.

6 Experimental inputs and gauge couplings at M_Z

In this section we compute the Standard Model gauge couplings explicitly from experimental inputs at the Z -pole and prepare them for comparison with the internal overlap formulation.

6.1 Electroweak inputs

We use the following $\overline{\text{MS}}$ -scheme inputs at $\mu = M_Z$:

$$\alpha(M_Z)^{-1} = 127.95, \quad \sin^2 \theta_W(M_Z) = 0.23122, \quad \alpha_s(M_Z) = 0.1179. \quad (6.1)$$

These declared illustrative inputs are retained for the elementary conversion below; the selected multi-loop profile is specified separately.

6.2 Definition of gauge couplings

Define the electromagnetic coupling

$$e := \sqrt{4\pi\alpha(M_Z)}. \quad (6.2)$$

Then the electroweak couplings are

$$g_2 = \frac{e}{\sin \theta_W}, \quad g' = \frac{e}{\cos \theta_W}, \quad (6.3)$$

and the GUT-normalized hypercharge coupling is

$$g_1 = \sqrt{\frac{5}{3}} g'. \quad (6.4)$$

The strong coupling is

$$g_3 = \sqrt{4\pi\alpha_s(M_Z)}. \quad (6.5)$$

6.3 Explicit numerical evaluation

From (6.1):

$$\alpha(M_Z) \approx 0.007816, \quad e = \sqrt{4\pi\alpha} \approx 0.3135.$$

Also,

$$\sin \theta_W = \sqrt{0.23122} \approx 0.4809, \quad \cos \theta_W \approx 0.8768.$$

Thus,

$$g_2(M_Z) = \frac{0.3135}{0.4809} \approx 0.6517,$$

$$g'(M_Z) = \frac{0.3135}{0.8768} \approx 0.3576,$$

$$g_1(M_Z) = \sqrt{\frac{5}{3}} \times 0.3576 \approx 0.4614.$$

For QCD,

$$g_3(M_Z) = \sqrt{4\pi \times 0.1179} \approx 1.2172.$$

Scale	g_1 (GUT)	g_2	g_3
M_Z	0.4614	0.6517	1.2172

Table 6.1: Gauge couplings at $\mu = M_Z$ computed from (6.1).

7 Selected multi-loop common-scheme transport

The former one-loop evolution to 5 TeV is not used in this revision. In particular, the Standard Model equations do not place the $g_1 = g_2$ crossing there. We instead use the selected SMDR v1.3 transport, which maps fifteen measured source coordinates to the full non-decoupled Standard Model in the tadpole-free pure $\overline{\text{MS}}$ scheme at

$$Q = M_t = 172.5590883453979 \text{ GeV}.$$

This is a scheme and comparison scale, not an MTT coherence scale.

The accepted gauge rows are

$$g_Y(Q) = 0.3585945042 \pm 0.0000307251, \tag{7.1}$$

$$g_2(Q) = 0.6475986708 \pm 0.0000287665, \tag{7.2}$$

$$g_3(Q) = 1.163427409 \pm 0.004036156. \tag{7.3}$$

With the paper's GUT normalization,

$$g_1(Q) = \sqrt{\frac{5}{3}} g_Y(Q) = 0.4629435143 \pm 0.0000396660.$$

These rows are outputs of a common multi-loop matching/running map. Their covariance is propagated from the declared diagonal measured-input profile; an official joint likelihood spanning all fifteen source coordinates is not claimed.

8 Numerical Θ -profile targets from gauge data

Under the coupling-overlap relation already proved conditionally above,

$$\frac{I_b}{I_a} = \left(\frac{g_a}{g_b} \right)^2.$$

Consequently the selected common-scheme profile gives

$$\boxed{\frac{I_2}{I_1} = 0.5110273 \pm 0.0001231, \quad \frac{I_3}{I_1} = 0.158335 \pm 0.001098.} \quad (8.1)$$

The propagated covariance of the two ratios is -6.1892×10^{-9} , corresponding to correlation -0.04578 .

These are experimentally anchored profile targets. A geometry adjusted to reproduce them is a calibrated realization. A held-out prediction would require fixing that geometry without these gauge rows and then computing an observable not used in source, branch, scale, or model selection.

8.1 Worked example: reading the target pair

As a concrete foothold, ignore the common normalization g_{10} and set $I_1 = 1$ only for reading the ratios. Equation (8.1) then says

$$I_2 \simeq 0.5110, \quad I_3 \simeq 0.1583.$$

The same information can be written as coupling ratios,

$$\frac{g_2}{g_1} = \frac{1}{\sqrt{I_2/I_1}} \simeq 1.3989, \quad \frac{g_3}{g_1} = \frac{1}{\sqrt{I_3/I_1}} \simeq 2.5131.$$

Nothing has been predicted in this rewrite: it is the same measured profile expressed in geometric coordinates. The useful test comes next, when a specified internal ansatz must realize both numbers without violating its gap and positivity conditions.

9 Transition to the geometric Θ -problem

The next step is to determine whether the internal geometry of MTT admits overlap integrals I_1, I_2, I_3 satisfying the above ratios while maintaining the declared baseline spectral floor 0.25.

This problem is addressed in the next section.

10 Baseline internal geometry and spectral gap bounds

We now specify the auxiliary operator models used for the Route-A gap check and collect their assumed spectral inequalities.

10.1 Auxiliary metric supports

The three vertical operators are tested on effective supports Σ_n :

- Σ_1 is an auxiliary circle,
- Σ_2 is a two-dimensional lens-base model,
- Σ_3 is a compact nilmanifold model.

The common $U(1)$ phase/holonomy line acts across all three supports and is counted once. The notation does not assert three products $S_{\text{cen}}^1 \times \Sigma_n$ or identify these supports with coordinate factors of the selected q79 compactification.

The auxiliary metric ansatz has parameters:

- R_1 – radius of Σ_1 ,
- R_{lens} – effective lens-base radius,
- f_2 – lens-base scale factor,
- h_0 – Cheeger lower-bound parameter for the nil model.

10.2 Spectral gap bounds

The following are model-specific lower bounds on the smallest nonzero eigenvalues of the auxiliary vertical Laplacians.

Circle factor Σ_1 . For the scalar Laplacian on the auxiliary circle of radius R_1 ,

$$\lambda_{\Sigma_1} = \frac{1}{R_1^2}. \quad (10.1)$$

Lens factor Σ_2 . For the lens sheet with effective radius $f_2 R_{\text{lens}}$,

$$\lambda_{\text{lens}} \geq \frac{2}{(f_2 R_{\text{lens}})^2}. \quad (10.2)$$

Nil factor Σ_3 . For the nil leaf, the spectral gap is controlled by the Cheeger constant:

$$h \geq h_0 > 0 \quad \Rightarrow \quad \lambda_{\text{nil}} \geq \frac{h_0^2}{4}. \quad (10.3)$$

10.3 Baseline numerical values

In the worked numerical baseline of the Foundation,

$$h_0 = 1 \quad \Rightarrow \quad \lambda_{\text{nil}} \geq \frac{1}{4} = 0.25. \quad (10.4)$$

We enforce the *baseline lower bound* $\lambda_* \geq 0.25$ using $h_0 = 1$. This sets a conservative admissibility floor; explicit realizations of the nil sector may exceed this bound and need not saturate it.

11 Geometric overlap model

We now introduce a minimal, computable auxiliary coefficient model. It is not obtained by multiplying three sector supports or by appending the shared circle to each support. Instead, all spectator integrations, mode normalizations, and gauge-kinetic weights are absorbed into three positive coefficients.

11.1 Auxiliary coefficient ansatz

At the Route-A calibration tier, parameterize the coefficients as

$$I_1 = 2\pi R_1, \quad (11.1)$$

$$I_2 = \kappa_\ell (f_2 R_{\text{lens}})^2, \quad (11.2)$$

$$I_3 = \kappa_n \mathcal{S}_n, \quad (11.3)$$

This is an ansatz for the weighted coefficients defined above, not a theorem equating the norm of an L^2 -normalized mode with a geometric volume. A selected compactification must emit the corresponding weights and representatives on one X_6 before these coefficients become source-derived. Here:

- κ_ℓ is the unit-area normalization of the lens sheet,
- $\kappa_n \mathcal{S}_n$ is the effective nil overlap weight,
- $\mathcal{S}_n$ is not assumed equal to a simple metric area.

This separation is required because the nil spectral gap (10.3) is controlled by h_0 , not directly by metric scale.

12 Solving the Θ -system

We now solve the overlap constraints (8.1) in the displayed auxiliary coefficient ansatz, together with the assumed spectral bounds. Decimal coefficients below are rounded representations of the expressions containing π , not additional exact input identities.

12.1 Overlap constraints

From (8.1) and (11.1)–(11.3):

$$\frac{I_2}{I_1} = \frac{\kappa_\ell (f_2 R_{\text{lens}})^2}{2\pi R_1} = 0.5110273, \quad (12.1)$$

$$\frac{I_3}{I_1} = \frac{\kappa_n \mathcal{S}_n}{2\pi R_1} = 0.158335. \quad (12.2)$$

Solving (12.1) gives

$$(f_2 R_{\text{lens}})^2 = \frac{0.5110273 \cdot 2\pi}{\kappa_\ell} R_1 = \frac{3.210879}{\kappa_\ell} R_1, \quad (12.3)$$

hence

$$R_{\text{lens}} = \frac{\sqrt{(3.210879/\kappa_\ell) R_1}}{f_2}. \quad (12.4)$$

From (12.2):

$$\kappa_n \mathcal{S}_n = 0.158335 \cdot 2\pi R_1 = 0.994849 R_1. \quad (12.5)$$

12.2 Imposing gap inequalities

We require:

$$\lambda_{\Sigma_1} \geq 0.25, \quad (12.6)$$

$$\lambda_{\text{lens}} \geq 0.25, \quad (12.7)$$

$$\lambda_{\text{nil}} \geq 0.25. \quad (12.8)$$

Nil. Set $h_0 = 1$, so (10.3) gives $\lambda_{\text{nil}} \geq 0.25$.

Circle. From (10.1) and (12.6),

$$\frac{1}{R_1^2} \geq 0.25 \quad \Rightarrow \quad R_1 \leq 2.$$

Lens. From (10.2) and (12.7),

$$\frac{2}{(f_2 R_{\text{lens}})^2} \geq 0.25 \quad \Rightarrow \quad (f_2 R_{\text{lens}})^2 \leq 8.$$

Substitute (12.3):

$$\frac{3.210879}{\kappa_\ell} R_1 \leq 8 \quad \Rightarrow \quad R_1 \leq \frac{8\kappa_\ell}{3.210879} \approx 2.492 \kappa_\ell.$$

12.3 Existence theorem

Theorem 12.1 (Existence of an auxiliary calibrated Θ -region). *At the selected common-scheme profile point $Q = M_t$, choose*

$$h_0 = 1, \quad \kappa_\ell = 1, \quad f_2 \geq 1.$$

Then for every $R_1 \in (0, 2]$ there exists a choice of R_{lens} and $\mathcal{S}_n$ given by (12.4) and (12.5) such that:

1. *the gauge overlap constraints $\frac{I_2}{I_1} = 0.5110273$ and $\frac{I_3}{I_1} = 0.158335$ are satisfied;*
2. *the spectral gap inequalities $\lambda_{\Sigma_1}, \lambda_{\text{lens}}, \lambda_{\text{nil}} \geq 0.25$ hold;*
3. *the guaranteed spectral floor remains $\lambda_* \geq 0.25$.*

Proof. Let $R_1 \in (0, 2]$. Then $\lambda_{\Sigma_1} \geq 0.25$. With $\kappa_\ell = 1$ and $f_2 \geq 1$, (12.4) implies $(f_2 R_{\text{lens}})^2 = 3.210879 R_1 \leq 6.421758 < 8$, hence $\lambda_{\text{lens}} > 0.25$. With $h_0 = 1$, $\lambda_{\text{nil}} \geq 0.25$. The overlap constraints are satisfied by construction. $\square$

13 Interpretation

The existence theorem shows only that the measured Standard Model gauge profile can be represented inside this auxiliary coefficient ansatz without violating its assumed dimensionless gap inequalities. The choice $h_0 = 1$ sets the declared nil lower-bound certificate in this ansatz; it does not prove that the selected q79 geometry has this value or that the nil lane physically controls the onset of noncoherent behavior.

Remark 13.1. The quantities κ_n and $\mathcal{S}_n$ encode nontrivial nilmanifold structure and are not free once explicit harmonic representatives are specified.

Remark 13.2 (Coherent sector stability). All results in this work are conditional on the stability of the coherent sector. Instability removes this paper's controlled coherent truncation and its error estimates. It does not imply the absence of physical interactions, ordinary measurements, records or probabilities in other descriptions.

14 Massless gauge-sector normalization via the twistor corner

The remaining technical ambiguity in the Θ -closure analysis concerns the explicit definition and normalization of the nonabelian gauge-kinetic coefficients I_2 and I_3 associated with the $SU(2)$ and $SU(3)$ sectors. In this section we state a well-typed conditional definition in the high-coherence (twistor) corner of Modal Triplet Theory.

Remark 14.1 (Twistor corner as a computational device). In this paper the twistor corner is used solely as a *computational normalization regime* for the massless coherent gauge sector, providing canonical harmonic representatives and gap-controlled error bounds. No claim is made that Nature resides exactly in this corner, nor that twistor space is fundamental.

14.1 High-coherence regime and admissible truncation

The twistor corner used here is a conditional high-coherence regime [6] in which:

1. the spectral gap λ_Q separating coherent and noncoherent modes is parametrically large;
2. the effective generator admits a Schur–Feshbach reduction onto a finite-dimensional coherent subspace;
3. truncation errors are bounded by explicit operator-norm estimates of order $O(\lambda_Q^{-1})$.

Assumption 14.2 (Twistor-corner admissibility). We assume that the Θ -matching scale μ_Θ lies within the high-coherence regime for the massless gauge sector, so that the Schur–Feshbach reduction is valid with remainder bounded by $O(\lambda_Q^{-1})$.

Remark 14.3. This assumption invokes no ultraviolet completion and no string-theoretic structure; it relies on the declared coherence, gap and truncation hypotheses.

14.2 Canonical definition of the massless gauge subspace

Let $\mathcal{H}$ denote the full internal Hilbert space of modal fields. Define $\mathcal{H}_0 \subset \mathcal{H}$ as the coherent massless gauge subspace selected by the twistor corner, characterized by:

1. self-duality of the Yang–Mills curvature;
2. vanishing effective mass under the reduced generator;
3. holomorphic encoding via the Ward correspondence.

Definition 14.4 (Twistor-corner massless sector). $\mathcal{H}_0$ is defined as the image of the coherent projector P_0 associated with the self-dual Yang–Mills corner of the MTT configuration space.

This definition is relative to the supplied operator, metric, domain and isolated spectral slab; it does not select those data physically.

14.3 Schur–Feshbach reduction and error control

Let L be the full internal generator and write

$$L = \begin{pmatrix} P_0 L P_0 & P_0 L Q \\ Q L P_0 & Q L Q \end{pmatrix}, \quad Q = 1 - P_0.$$

The effective generator on $\mathcal{H}_0$ is given by the Schur–Feshbach map

$$L_{\text{eff}} = P_0 L P_0 - P_0 L Q (Q L Q)^{-1} Q L P_0.$$

Under the stated complementary inverse and off-diagonal operator bounds, the Schur–Feshbach correction obeys

$$\|P_0 L Q (Q L Q)^{-1} Q L P_0\| \leq C \lambda_Q^{-1},$$

for some constant C independent of infrared data.

14.4 Harmonic representatives and normalization

Definition 14.5 (Twistor-corner harmonic representatives). For $a = 2, 3$, fix a cohomology class together with its period normalization, the internal metric, and a gauge condition. Let $\omega_a^{(0)}$ denote its harmonic representative in $\mathcal{H}_0$. Hodge theory fixes the harmonic representative of the fixed class; the period condition, rather than unit L^2 normalization, fixes its scale.

Remark 14.6. The abelian overlap I_1 remains fixed by the explicit S_{cen}^1 normalization $I_1 = 2\pi R_1$ in (11.1); the twistor construction does not replace that declared convention.

14.5 Leading-order overlap integrals

Definition 14.7 (Leading-order overlaps). The leading-order nonabelian overlap integrals are defined by

$$I_a^{(0)} := \int_{B_a} w_a^{(0)} \langle \omega_a^{(0)}, \omega_a^{(0)} \rangle d\mu_{B_a}, \quad a = 2, 3.$$

These depend on the massless coherent geometry, the fixed periods, and the gauge-kinetic weights. The twistor representation alone does not select those normalization data.

14.6 Control of overlap deviations

Let I_a denote the full weighted gauge-kinetic overlaps defined above.

Lemma 14.8 (Conditional overlap stability under coherent truncation). *Assume the full and leading projectors, period-normalized representatives, weights, and measures obey uniform perturbation estimates of order λ_Q^{-1} in norms controlling the displayed integrals. Then there exist constants $C_a > 0$ such that*

$$|I_a - I_a^{(0)}| \leq C_a \lambda_Q^{-1}, \quad a = 2, 3.$$

Proof. Expand the difference of the two weighted quadratic integrals into the representative, weight, and measure perturbations. Cauchy–Schwarz and the assumed uniform bounds control each term by its corresponding $O(\lambda_Q^{-1})$ estimate. Their finite sum gives the stated constant C_a . The Schur–Feshbach bound alone would not supply all of these coefficient estimates. $\square$

14.7 Twistor-corner Θ -closure at leading order

Combining the above with the experimentally extracted targets at the selected common-scheme point $Q = M_t$:

$$\frac{I_2}{I_1} = 0.5110273, \quad \frac{I_3}{I_1} = 0.158335,$$

we obtain:

Theorem 14.9 (Twistor-corner Θ -closure condition). *If the twistor-corner admissibility assumption and the preceding projector, representative, weight and measure perturbation bounds hold at μ_Θ , then Θ -closure at leading order reduces to the computable conditions*

$$\frac{I_2^{(0)}}{I_1} = 0.5110273, \quad \frac{I_3^{(0)}}{I_1} = 0.158335,$$

with controlled corrections of order $O(\lambda_Q^{-1})$.

Remark 14.10. Failure of these relations falsifies either the baseline internal geometry or the twistor-corner admissibility hypothesis.

14.8 Scope

This section does not claim that the twistor corner describes the full physical gauge sector. It provides a conditional computational corner in which I_2 and I_3 are well defined after the metric, periods, weights, and perturbation bounds have been supplied.

15 Scale separation and withdrawn legacy calibration

The scale $Q = M_t$ used above is a renormalization/matching coordinate. It does not determine the physical internal gap, compactification radius, proper-time filter, Planck scale, Hubble scale, or primordial tensor amplitude. The former identifications

$$Q \sim E_{\text{gap,min}} \sim \tau_0^{-1/2}$$

were additional calibrations tied to the invalid 4.2–5 TeV crossing and are withdrawn.

The dimensionless spectral inequalities in the preceding sections remain conditional geometric statements. Converting them to SI or external energy units requires an independently selected action normalization and a theorem relating the internal operator spectrum to physical propagator poles or response scales. No cosmological or quantum-gravity bound is inferred in this paper.

16 Falsifiability and status

At the profile tier, the direct test is whether a specified internal geometry, with its measure and harmonic normalization fixed independently of the gauge targets, emits the two ratios in (8.1) within the propagated covariance. If the geometry is fitted to those ratios, the result is instead a realization and round-trip consistency check.

The old cosmological and Gaussian quantum-gravity discriminators are removed because their physical scale identification is not derived here. The current broader MTT repository closes embedded renormalized-Standard-Model equivalence at the adopted one-shared-physical-primitive/profile standard. That successor does not turn the present overlap targets into zero-knob predictions and does not derive standard perturbative quantization from MTT.

17 Conclusion

The coupling–overlap formula survives as a conditional dimensional-reduction identity. Its numerical implementation is now placed in one selected multi-loop common scheme at $Q = M_t$, yielding

$$I_2/I_1 = 0.5110273 \pm 0.0001231, \quad I_3/I_1 = 0.158335 \pm 0.001098.$$

The full covariance provenance is retained, and the absence of a public joint fifteen-coordinate likelihood is stated explicitly.

The former 5 TeV crossing, the geometry calibrated to its ratios, and the identification of that scale with an internal gap, proper-time cutoff, or cosmological scale are not results of this revision. Revised Papers II–V retain the corresponding calibrated, conditional and round-trip roles. This paper therefore supplies a reproducible profile target and a precise geometric test, not a unique first-principles prediction of gauge couplings or internal geometry.

18 Selected-source results and this paper’s boundary

The eta9 imports are owned contextually by *Flux Compactifications in Heterotic String Theory*, in the sections on the selected source and the integral comparison, and completed local tests and the global endpoint. Gate 1 closes the original-Jacobian campaign at all 30 groups and 225 selected columns [7]. This is an exact finite source calculation, not the global integral meridian, the 248-coordinate readout or the analytic Deligne class β_C .

The completed three-cycle transport has independent cycles with Gram matrix $-2I_3$ and zero detected affine pairings [8]. Its scope is a non-detecting subsystem: it does not prove global triviality of the twist, and rerunning that same subsystem is not the missing detection theorem. Neither this conclusion nor the finite gate identifies an auxiliary Circle–Lens–Nil model with the physical q79 topology.

The large-gauge result is owned by *Cohesive Closure Repair and Its Hodge, Kernel, and Projection Shadows*, in its subsection on the rank-zero large-gauge kernel. It gives an injective immersion $b_{K3} : \mathbb{R}^2 \rightarrow \mathbb{T}^{20}$ with dense nonclosed image [9]. Rank zero counts periodic gauge identifications, not source coordinates or physical modes. It removes a kernel ambiguity without requiring numerical periods, but does not provide kinetic normalization, the selected shared-circle action or the unrelated eta9 affine lift.

For Paper I, none of these results selects the period-normalized modes, gauge-kinetic weights or action scale that would emit the two measured overlap targets. A rank-zero gauge kernel is not zero-input gauge prediction. The selected eta9 local source tube is also not a useful-width global transport certificate. The established one-shared-primitive Standard Model profile closure remains retained; the stricter source-independent overlap prediction is a different question.

The managed evidence block below retains its earlier frozen profile snapshot. Its historical source-status labels do not supersede the current exact, local and open boundaries just stated or reopen the retained hidden-HYM existence and shared-primitive Standard Model results.

Computational Evidence and Reproducibility

This paper uses the common-scheme precision transport, electroweak row, HYM certificate, and global profile audit to define and check its gauge-overlap targets. Flavor and neutral packets are neighboring profile context rather than inputs to the local overlap identity. The geometric source and zero-knob upgrade remain open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] Particle Data Group (PDG), Review of Particle Physics (declared electroweak inputs).
- [2] M. E. Machacek and M. T. Vaughn, “Two-loop renormalization group equations in a general quantum field theory,” Nucl. Phys. B222 (1983) 83–103.
- [3] T. Kato, *Perturbation Theory for Linear Operators*, Springer (1995).
- [4] P. Nero, *Modal Triplet Theory: Foundation* (MTT corpus).
- [5] P. Nero, *Modal Triplet Theory: Quantum Amplitudes from Modal Geometry* (MTT corpus).

- [6] P. Nero, *Twistor Encodings as High-Coherence Limits of Modal Triplet Theory* (MTT corpus).
- [7] P. Nero, *Selected eta9 original-Jacobian Gate 1 campaign* (2026). Frozen source at curated manifest commit `f141a20e a23c5c3f f19cc216 1c0e226e 29ade8a7`: [source artifact](#).
- [8] P. Nero, *Complete selected three-cycle non-detection decision* (2026). Frozen source at curated manifest commit `f141a20e a23c5c3f f19cc216 1c0e226e 29ade8a7`: [source artifact](#).
- [9] P. Nero, *Selected K3 rank-zero large-gauge kernel* (2026). Frozen source at curated manifest commit `f141a20e a23c5c3f f19cc216 1c0e226e 29ade8a7`: [source artifact](#).